

100: NEURAL NETWORK ACCELERATOR (GPU / NPU ARCHITECTURE)

102: PRIMARY ACCELERATOR MEMORY (HBM3e / VRAM)

104: Contiguous Dynamic KV-Cache Tensor Buffer [B, H, L, D_k]



Direct High-Speed Memory Bus (Zero Intermediate cudaMalloc Calls)

Bandwidth: up to 8.0 TB/s (NVIDIA Blackwell sm_120 / HBM3e)

110: INTEGRATED IN-PLACE MEMORY MANAGEMENT CONTROLLER

112: Associative Scoring & Characteristic Mapping Unit

- Tracks cumulative attention scores $S(k_i)$; partitions active vs evicted token slots

114: Idempotent Projection Engine

- Enforces $f(f(x)) = f(x)$; locks active tokens into stabilized attractor basins

116: In-Place Disjoint Cycle-Leader Permutation Engine

- $O(1)$ auxiliary registers; executes zero-copy cyclic orbit swaps without bitmasks

140: STREAMING MULTIPROCESSORS / TENSOR CORES (PARALLEL COMPUTE)

Vectorized Attention Matrix: $\text{Softmax}(Q * K^T / \sqrt{d}) * V$ on Compact Contiguous Memory